

Primary UD, EPR, 5 kV - 35 kV

Copper Conductor, EPR Insulated
Concentric Neutral, LDPE Jacket

Features

Low tension stripping compounds

True Triple extrusion system and closed handling raw materials system, to eliminate any contact with ambient, until extrusion of insulation and shields

Application

Underground primary residential and commercial distribution circuits

May be used in wet or dry locations, installed in underground ducts or direct burial

Standards

ICEA S-94-649:
Standard for concentric neutral rated cables 5 to 46 kV

AEIC CS8:
Specifications for extruded dielectric, shielded power cable, rated 5 to 46 kV

SPECIFICATIONS

Maximum operating voltage:

5 kV - 35 kV, 100% & 133% IL

Maximum conductor operation temperatures:

Wet and dry locations:

- Normal: 105°C
- Emergency: 140°C
- Short Circuit: 250°C

Engineering Information

Conductor: Soft-annealed uncoated copper compressed Class B stranding, or unilay-compressed per ASTM B 8. *On Request: Aluminum conductor and/or strand filled*

Sizes: 8 AWG - 1 000 kcmil. *On request larger sizes*

Conductor Shield: Semi conducting cross-linked polyethylene (XLPE). *On request, super-clean and super-smooth materials*

Insulation: Thermoset ethylene propylene rubber (EPR). *On request amorphous EPR*

Insulation Shield: Semi conducting cross-linked polyethylene (XLPE)

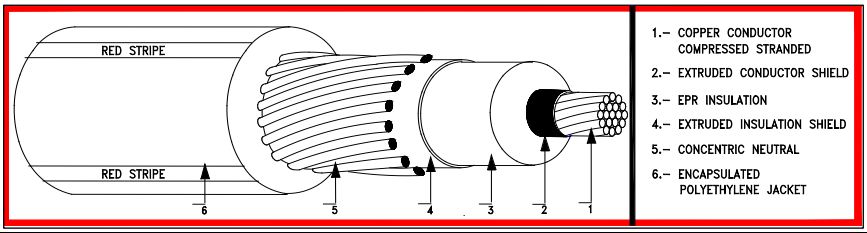
Concentric Neutral: Soft-annealed solid copper wires per ASTM B 3, helically applied and uniformly spaced. Full or 1/3 Neutral. *On request per customer specs,*

Jacket: Extruded to fill (Encapsulated) Black sunlight resistant low density polyethylene (LDPE), with three Red Stripes. *On request: Overlying (Sleeve) Black polyvinyl chloride (PVC), or semiconducting LDPE*

Configuration Options*: On Request, Triplex or Paralleled configurations*

CME wire and cable

Primary UD, EPR, 15 kV Copper Conductor, EPR Insulated, Concentric Neutral, LDPE Jacket



1/3 NEUTRAL — *15 kV 100% INSULATION LEVEL*

Size AWG or kcmil	Number of Strands	Insulation Thickness Mils	1/3 Neutral No. of wires Mils	1/3 Neutral Size AWG Mils	Jacket Thickness Mils	Approximate Outside Diameter Inches	Approximate Net Weight Lbs/Mft
2	7	175	6	14	50	0.98	606
1	19	175	7	14	50	1.02	691
1/0	19	175	9	14	50	1.06	805
2/0	19	175	11	14	50	1.10	939
3/0	19	175	14	14	50	1.15	1110
4/0	19	175	18	14	50	1.21	1325
250	37	175	13	12	50	1.30	1534
350	37	175	18	12	50	1.42	2023
500	37	175	17	10	50	1.60	2780
750	61	175	20	9	80	1.87	4021
1000	61	175	26	9	80	2.05	5174

FULL NEUTRAL — *15 kV 100% INSULATION LEVEL*

Size AWG or kcmil	Number of Strands	Insulation Thickness Mils	Full Neutral No. of wires Mils	Full Neutral Size AWG Mils	Jacket Thickness Mils	Approximate Outside Diameter Inches	Approximate Net Weight Lbs/Mft
2	7	175	16	14	50	0.98	726
1	19	175	13	12	50	1.05	878
1/0	19	175	16	12	50	1.09	1026
2/0	19	175	13	10	50	1.18	1257
3/0	19	175	16	10	50	1.23	1486
4/0	19	175	16	9	50	1.31	1804
250	37	175	25	10	50	1.34	2082
350	37	175	22	8	50	1.52	2834
500	37	175	31	8	80	1.71	3886

The above data are approximate and subject to normal manufacturing tolerances.

1/3 NEUTRAL — *15 kV 133% INSULATION LEVEL*

Size AWG or kcmil	Number of Strands	Insulation Thickness Mils	1/3 Neutral No. of wires Mils	1/3 Neutral Size AWG Mils	Jacket Thickness Mils	Approximate Outside Diameter Inches	Approximate Net Weight Lbs/Mft
2	7	220	6	14	50	1.07	680
1	19	220	7	14	50	1.11	768
1/0	19	220	9	14	50	1.15	885
2/0	19	220	11	14	50	1.20	1022
3/0	19	220	14	14	50	1.25	1197
4/0	19	220	18	14	50	1.30	1417
250	37	220	13	12	50	1.41	1656
350	37	220	18	12	50	1.52	2130
500	37	220	17	10	80	1.75	2967
750	61	220	20	9	80	1.97	4160
1000	61	220	26	9	80	2.15	5327

FULL NEUTRAL							*15 kV 133% INSULATION LEVEL*
Size AWG or kcmil	Number of Strands	Insulation Thickness	Full Neutral No. of wires	Full Neutral Size AWG	Jacket Thickness	Approximate Outside Diameter	Approximate Net Weight
		Mils	Mils	Mils	Mils	Inches	Lbs/Mft
2	7	220	16	14	50	1.07	800
1	19	220	13	12	50	1.14	957
1/0	19	220	16	12	50	1.18	1108
2/0	19	220	13	10	50	1.27	1345
3/0	19	220	16	10	50	1.32	1578
4/0	19	220	16	9	50	1.40	1901
250	37	220	25	10	50	1.46	2208
350	37	220	22	8	50	1.61	2946
500	37	220	31	8	80	1.80	4012

The above data are approximate and subject to normal manufacturing tolerances.

5 kV - 15 kV 100% or 133% Insulation Level

Ampacity

Size AWG or kcmil	† Ampacity Full Neutral	† Ampacity 1/3 Neutral
2	160	160
1	185	187
1/0	210	212
2/0	238	241
3/0	268	273
4/0	300	309
250	-	336
350	-	398
500	-	476
750	-	547
1000	-	620

Rating Voltage	Minimum Size AWG
8 kV	6
15 kV	2
25 kV	1
35 kV	1/0

Three shielded single insulated copper conductors in buried ducts, single circuit, 25°C ambient earth temperature, 75% load factor, 90 RHO and 90°C conductor temperature (IEEE Std. 835).